

Deliverable D - Technical Writeup

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1. Problem framing & assumptions violated

We treat this as **lending under selective labels and partial observability**, not generic classification. The data spans **85,340 train, 4,489 validation, and 8,817 test** applications (51,722 and 2,551 carry repayment labels in train and validation; test is unlabeled). A repayment label exists *iff* the prior lender approved the application (`prior_decision = 1`), and every approved loan in the data has matured (no right-censoring): the labeled region is a non-random, policy-selected slice of applicant space. The standard assumptions break:

- **Labels-for-all is false.** Declined applicants have no outcome, and the missingness is driven by the risk we are predicting (informative / MNAR at the policy level), not ignorable noise.
- **IID / no-shift is false.** Default rate rises 17.4% (train, Jan 2024–Jun 2025) to 20.6% (validation). The **test window equals the validation window** (2025-06-30 to 2025-09-28) and is strictly later than train, so we must deploy under the later, higher-default regime — not train's.
- **prior_decision is degenerate as a feature** (constant on every labeled row): excluded from the PD feature set, repurposed as the *target* of an approval-propensity model used for overlap diagnostics.
- **Availability is selective too.** Bank-feed fields are null for ~35.7% of applicants (no linked feed); we model that missingness as signal rather than imputing it away.

2. Methodology

Features. We drop the six outcome columns, identifiers, the raw timestamp, and `prior_decision`; keep all raw numerics (including the always-observed `prior_underwriter_score`), six low-cardinality categoricals passed natively to LightGBM, engineered leverage/affordability ratios, and four binary missingness indicators. LightGBM consumes NaN natively, preserving informative missingness.

PD model. A 5-member bagged LightGBM ensemble, **isotonically calibrated on 5-fold out-of-fold** predictions; the per-row logit-scale spread across members is retained as epistemic uncertainty. Because the operational era equals validation and post-dates train, we fit a 2-parameter logit-linear (Platt) map on validation outcomes (slope 0.972, intercept 0.123) inside the predictor. Its value is level, not rank: uncalibrated mean PD 0.1853 under-predicts the deployment rate (0.2062); recalibration corrects it to 0.2061 (ECE 0.0281 to 0.0246). **All of A, B, C use deployment-era PDs.**

Funding policy (A). Expected profit per dollar is $(1-\text{PD}) \cdot r_{\text{perf}} - \text{PD} \cdot \text{LGD}$, with **LGD = 0.9065** (recoveries over 9024 defaulters) and $r_{\text{perf}} = 0.0875$ (spec), giving an **analytic break-even PD of 0.0881**. We deploy **PD* = 0.0890**, a hybrid (break-even blended 50/50 with a validation-profit-tuned cutoff; the two nearly coincide). On validation this is the only profitable rule tested: **0.0178/\$ at a 7.0% approved-bad-rate**, versus a naive $\text{PD} < 0.20$ cutoff at **-0.0253/\$** (bad rate 11.4%, above break-even). The high LGD makes the break-even cutoff binding. The policy funds 15.6% of applicants at an approved-book mean PD of 6.7%.

Cohort trajectories (B). We factor *how many* default from *when*: $\text{CDR}(c, w) = M(c) \cdot G(w)$, with $M(c)$ the funded-book mean PD for cohort c and $G(w)$ the empirical fraction of eventual defaults realized by week w from 9024 matured defaulters. G is flat over weeks 9–12 then jumps at week 13 — the day-90 outstanding-balance rule firing a deterministic mass, which we keep. Trajectories are forced monotone by a running maximum. Timing is era-stable (train-vs-validation max $|G \text{ shift}| = 0.034$); per-cohort G is rejected as too sparse (median 40 defaulters/cohort). Every cohort is funded (≥ 128 approvals).

3. Causal reasoning & counterfactual methodology

Observational vs. interventional. A answers $P(\text{default} \mid X = x)$ — risk for an applicant as observed. C answers $P(\text{default} \mid \text{do}(\text{feature} = v))$ — risk if we externally set a feature and let its consequences

follow. Different estimands, handled differently.

Identification. We do not posit a full structural causal model; we use the calibrated PD regression as a response surface and transport it to the interventional query under **conditional ignorability** given the observed covariates — crucially including `prior_underwriter_score`, populated for 100% of rows, which absorbs the private information the prior underwriter selected on. Ignorability is stated as an assumption; we pay for its fragility with wider intervals, not hidden confidence.

Defending the drivers to a regulator. An ablation shows removing `prior_underwriter_score` moves validation AUC by only -0.0027 (within noise): the model does **not** lean on the opaque vendor score — its working drivers are observable economics (loan size, revenue, utilization, debt). We retain the score purely to absorb the selection mechanism. The table below shows the model's PD responds most to a genuine lever (`requested_amount`) and to financial/credit health, and barely to identity, prior-history, or proxy features — exactly where we also widen uncertainty most. Risk tracks economics, not identity.

Feature group (do-target)	Queries	Mean PD move	Mean width	Causal stance
actionable terms	40	0.084	0.197	lever (full)
financial health	281	0.043	0.236	interventional
credit bureau	243	0.034	0.217	interventional
bank feed behavior	24	0.004	0.310	structural+cascade
prior history	29	0.002	0.320	associative
downstream collider	70	0.006	0.308	collider/proxy
platform engagement	39	0.010	0.345	proxy
identity context proxy	174	0.006	0.325	structural/proxy

Point estimates use full model perturbation for every group. Because we cannot sign the confounding bias, the model conditional remains our best point estimate even for weakly-identified features; we encode lower credence as a wider interval (logit-scale extra term 0.18 for the clean lever up to 0.42 for structural/proxy), not as an ad-hoc shrink that would itself be an unjustified causal claim. **Dependency-aware propagation:** changing `requested_amount` or observed revenue recomputes `requested_amount_to_observed_revenue`; `do(has_linked_bank_feed = 0)` sets all bank-derived signals and that ratio to unobserved, so the applicant is priced through the missingness channel as in production — freezing those descendants would be incoherent.

Overlap / selective-labels stress test. Approval-propensity AUC is 0.7591 with $p(\text{approve})$ in 0.074–0.984 (no positivity violation; declined applicants are not pure extrapolation). Inverse-propensity reweighting moves the portfolio-mean PD only 0.020 (0.2756 to 0.2553), so the elevated risk on historically declined applicants is covariate-driven, not an extrapolation artifact. We therefore widen intervals in thin-overlap tails but do not reweight the point estimates.

4. Calibration & uncertainty quantification

Metric (validation)	Value	Reading
Mean predicted PD vs. actual	0.2061 vs 0.2062	level-calibrated
Brier score (labeled)	0.1354	sharp + calibrated
AUC: 5-fold OOF / validation	0.7694 / 0.7484	rank power, mild era decay
90% interval coverage (15 bins)	93.3%	meets nominal 90%
Per-quintile coverage	5 / 5 quintiles	covered across the risk range

Construction. All PD intervals are built on the logit scale: $\text{width} = z \cdot \sqrt{\text{epistemic var} + \text{floor}^2 + \text{extra}^2}$, $z = 1.645$. The epistemic term is the ensemble logit spread; a floor (0.150) tuned on validation guarantees nominal coverage where the ensemble is over-confident; an extrapolation

multiplier grows with low approval-propensity and bank-feed missingness. Calibrated on validation labeled rows, the floor was the smallest value giving $\geq 90\%$ bin coverage (the width–coverage trade-off); realized bin coverage is 93.3% and the realized default rate sits inside the averaged band in all five PD quintiles, i.e. mildly conservative. Mean width is 0.128 for A. **C is wider by construction** (mean 0.262) via the per-group extra term, widest for structural and proxy interventions. B combines the model term, the binomial variance of $G(w)$, and a 12% era-transfer floor by the delta method.

5. Limitations & what we'd do differently

- **Ignorability is unfalsifiable here** — no instrument or exogenous shock for approval. With a policy change, regional score cutoff, or randomized-approval pilot we would target an identified effect (IV / regression-discontinuity) rather than selection-on-observables.
- **Structural counterfactuals are flagged sensitivities, not effects.** Setting sector or vintage in isolation has no real-world referent; we widen and label them, but a faithful treatment needs a structural model of how such attributes co-move with the rest of the application.
- **One observed era beyond train.** G is from train-era defaulters; cohort mix and day-90 behavior may drift. We would refit G on deployment vintages as they mature and use calibration drift as a re-Platt-scaling trigger.
- **LGD is constant (0.9065).** Recovery varies with applicant and amount; an LGD(X) model would sharpen the profit calculation. r_{perf} is fixed by the brief.
- **Intervals are ensemble- and floor-based and mildly conservative** (bin coverage 93.3%). We would add split-conformal prediction for finite-sample coverage, re-tune the floor for sharpness if the scorer rewards it, and replace the mean $\times G$ factorization with an explicit discrete-time hazard model for the day-90 mass.